J. Geomag. Geoelectr., **32**, Suppl. III, SIII 11–SIII 21, 1980

Early Palaeozoic Palaeomagnetism in South East Australia

Bruce R. GOLEBY

Research School of Earth Sciences, Australian National University, Canberra A.C.T., Australia

(Received February 25, 1980 ; Revised April 30, 1980)

The palaeomagnetism of 22 formations from the Cowra Trough and Molong High of South East Australia have been studied. Samples of Ordovician through Middle Devonian age were collected in three major areas, covering a time period crucial to the understanding of the Palaeozoic apparent polar wander path of Australia. Five magnetic components have been identified: 1) A recent field overprint, 2) a secondary component related to the break up of Australia and Antarctica, 3) a predominent Devonian secondary overprint associated with the major orogenic episode, 4) a weak Silurian overprint associated with an earlier orogenic episode, and 5) primary magnetic components ranging in age from the Ordovician to the Devonian. The directions obtained enable construction of a well defined polar wander path for South East Australia and Australia as a whole.

1. Introduction

Since the work of LUCK (1973) on the Early Palaeozoic of South East Australia, there have been three interpretations of the available pole positions for Australia. MCELHINNY and EMBLETON (1974) and EMBLETON *et al.* (1974) proposed a two plate tectonic model based on a difference of 90 degrees in the apparent pole path for the South Eastern part of the Tasman Geosyncline (as defined by PACKHAM, 1969) compared with the simplest interpretation of the pole path for the stable central Australian craton (Fig. 1A). Poles SV, MP, AV, and BG represent the Siluro-Devonian track of South East Australian sub-plate. Since the Late Devonian, the sub-plate was welded to the main craton producing a common track for the whole of Australia.

SCHMIDT and MORRIS (1977), by reversing the polarity of the Cambro-Ordovician poles constructed a single apparent polar wander path using all the polar data (Fig. 1B). With SCHMIDT and MORRIS's (1977) model the suggested Early Devonian age of magnetization of the Upper Ordovician to Upper Devonian Mereenie Sandstone (MS), and the very fast polar wander rates required for this interpretation have yet to be explained successfully. Two strong points are in favour, 1) all the reliable Australian poles are used, and fit stratigraphically in sequence. 2) The angular distance between the Ordovician and the Silurian sections requires less than 90 degrees movement.

The third interpretation involves constructing a sequence of three 'loops' (MOREL and IRVING, 1978, path 'Y,' Fig. 1C), one during the Cambro-Ordovician, another during the Siluro-Devonian, and a third during the Late Devonian. The Stairway Sandstone (SS;

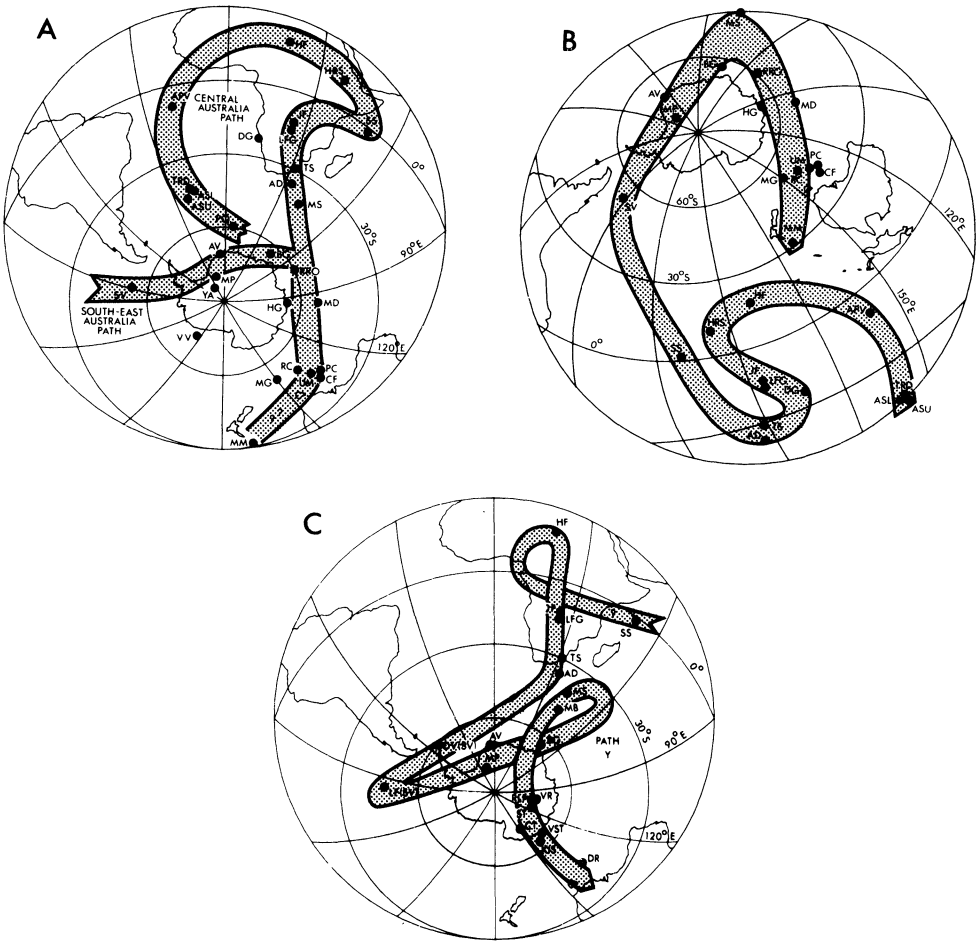


Fig. 1. A) McElhinny and Embleton's (1974) apparent polar wander path for Australia. The central Australian path defined by Early Palaeozoic poles connects with the South-eastern Australia path during the Late Devonian. B) Schmidt and Morris's (1977) path after adopting a reversed polarity for pre-Devonian poles. All the Australian poles fit stratigraphically in sequence. C) Morel and Irving's (1978) path 'Y'. The connection of the central Australian Ordovician poles to the South-eastern Australian Silurian poles is supported by new results, but the rest of the proposed path is inconsistent, both from stratigraphic relations and the validity of poles used. Polar linear projections. Poles are PQ, Pound Quartzite; ASU and ASL, Upper and Lower Arumbera Sdst; TRD, Todd River Dolomite; APV, Antrim Plateau Volc; HF, Hudson Fm; HRS, Hugh River Shale; SS, Stairway Sdst; JF, Jinduckin Fm; LFG, Lake Frome Group; DG, Dundas Group; TS, Tumblagooda Sdst; AD, Arooma Dam Seds; MS, Mereenie Sdst; SV, Silurian Volc; AV, Ainslie Vols; MP, Mugga Porphyry; BG, Bowning Group; RRO, Ross River Overprint; HG, Housetop Granite; MD, Mulga Downs; RC, Rocky Creek Cong; PC, Permo-carb. Volc; MG, Main Glacial Stage; CF, Currabubulla Fm; UM, Upper Marine Latites; MM, Milton Monzonite; DV(SV), Duro Volcs (Sil Volc); LF(SV), Laidlow Fm (Sil Volc); MB, Murrumbidgee Series; RLF, Redbeds-Lochiel Fm; ST, Tasmanian Sdst; VST, Tasmanian Volc and Seds; KIS, Kangaroo Island Seds; DR, Dotswood Redbeds; CT, Catombal Fm; and VR, Redbeds Vict.

EMBLETON, 1972) is Middle Ordovician (Upper Llanvirnian to Llandeilian; WELLS *et al.*, 1970) and the Hudson Formation (LUCK, 1972) is Middle Cambrian, yet MOREL and IRVING (1978) have their Cambro-Ordovician path passing through the Stairway Sandstone, then the Hudson Formation before connecting with the Cambro-Ordovician Jinduckin Formation (LUCK, 1972), and Lake Frome Group (EMBLETON and GIDDINGS, 1974) poles (Fig. 1C). The Upper Devonian 'loop' is based on selected poles assumed secondary components, yet valid primary poles have been left out and therefore this loop (Fig. 1C) is suspect. Secondly MOREL and IRVING's (1978) path requires movement through greater than 90° angular distance.

The purpose of this paper is to report briefly new palaeomagnetic results obtained from rock sequences of suitable age and location so as to resolve this ambiguity in the Early Palaeozoic polar wander path for Australia. Details on local geology, age relations, rock magnetic properties, and individual results will be presented elsewhere.

2. Geology of the Sampling Areas

Rock samples were collected from three areas within the Tasman Geosyncline of South East Australia. The Tasman Geosyncline has been subdivided into a series of

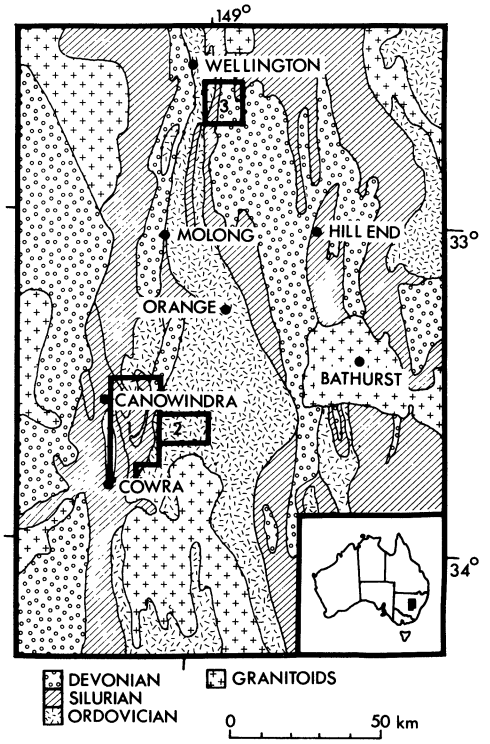


Fig. 2. Regional geology of the areas sampled. The area is part of the Molang High/Cowra Trough region of the Tasman Geosyncline of South-east Australia.

north-south trending tectonic provinces, two of which are the Molong High and Cowra Trough (Fig. 2).

The Molong High is dominantly Ordovician, with localized areas of Silurian and Devonian. The Ordovician sequences represent three episodes of tectonic activity: initial andesitic volcanism during the Early Ordovician, followed by a quiet period enabling the deposition of limestones, siltstones, and arenites during the Middle-Late Ordovician. This was followed by a further period of volcanic activity resulting in volcanic tuffs and lavas. Samples were collected stratigraphically through this sequence, and where possible, in the overlying Siluro-Devonian.

The Cowra Trough (Fig. 2) formed as a result of internal rifting within the Molong High during the Late Ordovician (SCHEIBNER, 1973). It is dominantly Silurian in age, with isolated areas extending into the Middle Devonian.

Four peaks in orogenic activity have been identified within the Tasman Geosyncline during the Early Palaeozoic, each resulting in either localized or widespread deformation at different times in different areas. The major folding in the areas sampled occurred during the Bowning Orogeny (Early Devonian), with minor folding, uplift and erosion occurring during the Early-Middle Silurian (Benambran Orogeny), the Middle Devonian (Tabberabberan Orogeny), and the Early Carboniferous (Kanimblan Orogeny).

3. Experimental Procedures

Samples were collected either as blocks or as drill cores orientated using a modified suncompass (EMBLETON and EDWARDS, 1973). Measurements of intensity and direction were made using either a 'complete results' DIGICO spinner magnetometer (MOLYNEUX, 1971) or a Super Conducting Technology (SCT) cryogenic magnetometer (GOREE and FULLER, 1976). Thermal and Alternating Field (AF) demagnetization procedures were carried out in air and in zero magnetic field using equipment similar to that described by MCELHINNY *et al.* (1971) and MCELHINNY (1966) respectively. Where specimens showed evidence of weathering in the form of a red staining, chemical demagnetization was carried out using a pressurized leaching apparatus (BUREK, 1971).

4. Results

The results will be discussed in terms of a particular time period, and hence, generalizations are made here for each time period. Details regarding each particular formation are to be presented elsewhere. Sampling localities were situated either on both limbs of the dominant fold, and/or from areas having a different bedding orientation. Application of GRAHAM's (1949) fold test on characteristic directions from these samples enabled positive recognization of the pre-folding direction. Each specimen was subjected to stepwise thermal or AF demagnetization up to a maximum of 675°C or 180 nT respectively to analyse the remanence components.

Fig. 3. Compilation of magnetic direction results. For the Ordovician and Silurian each point represents a mean of a minimum of three sites, while each point for the Devonian represents the mean of a single formation. Open circles, negative dips. Equal angle projection.

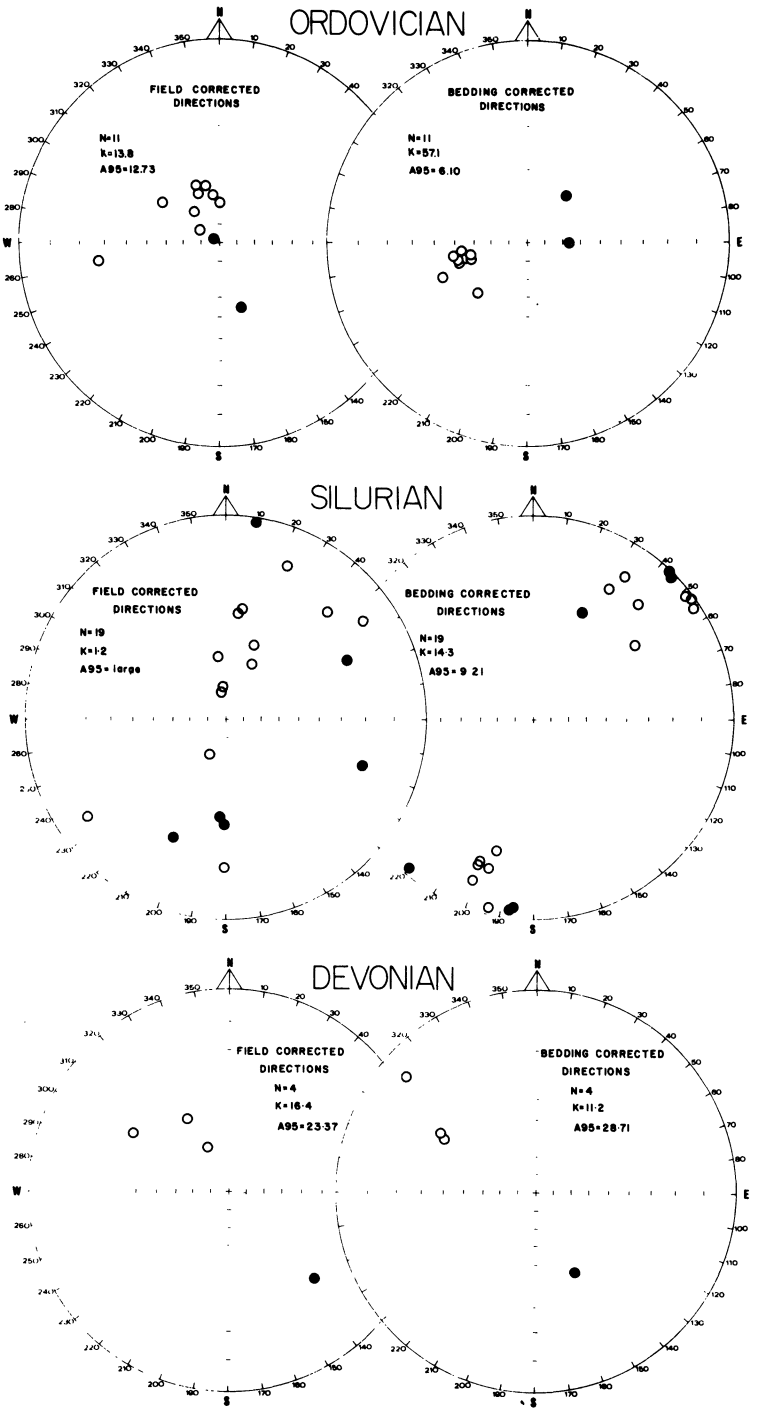

ORDOVICIAN
FIELD CORRECTED DIRECTIONS
N=11
k=13.8
A95=12.73
BEDDING CORRECTED DIRECTIONS
N=11
K=57.1
A95=6.10
SILURIAN
FIELD CORRECTED DIRECTIONS
N=19
K=1·2
A95= large
BEDDING CORRECTED DIRECTIONS
N=19
K=14·3
A95=9·21
DEVONIAN
FIELD CORRECTED DIRECTIONS
N=4
K=16·4
A95=23·37
BEDDING CORRECTED DIRECTIONS
N=4
K=11·2
A95=28·71

Table 1. Mean palaeomagentic data.

Age	Decl	Incl	k	A_{95}	Lat (°N)	Long (°E)	DP	DM
Middle-Late Ord.	251.1	−55.5	57.1	6.10	6.14	19.09	6.22	8.71
Early-Middle Sil.	45.9	−8.3	37.8	8.49	−38.23	34.55	4.31	8.56
Middle-Late Sil.	19.1	10.2	28.6	9.19	−47.34	357.48	4.71	9.30
Early-Middle Dev.	320.9	−24.4	11.2	28.71	−43.96	253.20	17.32	32.10
Secondary components								
Early-Middle Dev.	325.7	−12.8	26.7	9.53	−50.42	272.46	5.01	9.77
Mesozoic	20.6	−76.5	74.9	6.44	−56.62	132.78	11.08	11.94

Re-corrected AV and MP Poles

Formation name	Age	Old pole pos.		New pole pos.		
		Lat (°S)	Long (°E)	Lat (°S)	Long (°E)	
Mugga Porphyry	E. Ludlovian	80.	340.	63.2	7.9	
Ainslie Volc	Late Wenlock	71.	353.	72.3	341.2	for 5° dip
				67.6	285.3	for 35° dip

Eight Ordovician formations were investigated, three of which were completely overprinted. From the remaining Ordovician Formations a primary component and up to four secondary overprint directions were isolated. These were identified using ZIJDERVELD's (1967) orthogonal projections and the least squares method of data analysis. Each overprinting could be related to a major tectonic episode in South East Australia. The primary direction of magnetization was isolated within the temperature range 400°C–650°C. Field and bedding corrected directions for the Ordovician are shown in Fig. 3. After bedding correction the primary directions from the Middle-Late Ordovician Formations clustered about a mean direction of $D=251.1°$, $I=-55.5°$, with Fisher statistics $A_{95}=6.1°$ and $k=57.1$ (Table 1).

For a particular formation, if the magnetization of two (or more) limbs of folded rock strata are brought into agreement after correction for bedding, then the magnetization pre-dates the folding. MCELHINNY (1964) gives a statistical treatment of this test, and formations satisfying both criteria are said to have a positive fold test (age of magnetization pre-dates folding). The FISHER (1953) precision parameter k increases from 13.0 to 57.1 on restoring the Middle-Late Ordovician strata to the horizontal. This is significant at the 99% level according to MCELHINNY (1964). The age of magnetization is therefore pre-Benambran Orogeny, or Pre-Early-Middle Silurian in age. Both reversed and normal directions were present.

For the Silurian, eight out of a total of ten formations contained some remanence of the primary direction of magnetization. This characteristic primary direction was isolated somewhere in the temperature range 400°C–650°C (Fig. 3). Positive fold tests on the Silurian formations confirmed that the directions are primary. On bedding correction the resultant poles could be subdivided into two clusters. The oldest corresponds to an Early-Middle Silurian mean pole, the second to a Middle-Late Silurian mean pole (Fig. 4).

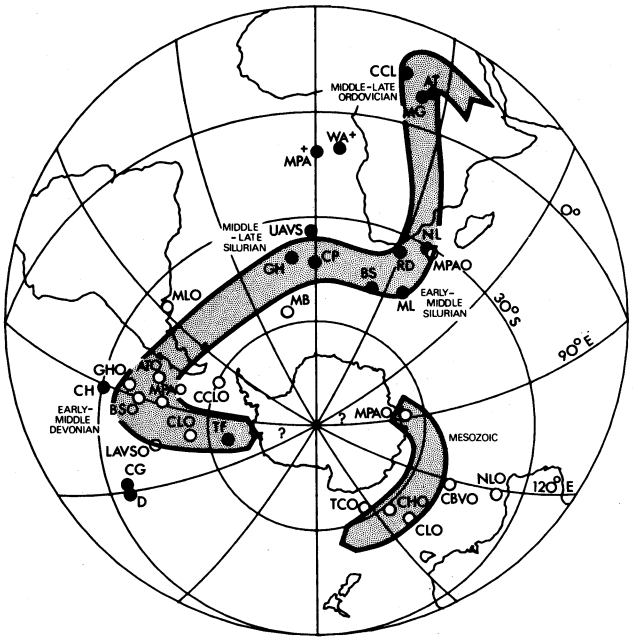


Fig. 4. Polar wander path constructed using new poles. The two poles marked with a cross are the only Early Ordovician poles obtained and complexities in the local structure may explain their position. The path is very well defined from the Ordovician to the Latest Silurian. The path between the Devonian and the Mesozoic may not be continuous, and for this reason is not connected. Solid circles, primary poles; Open circles, secondary poles. Mnemonic's used are listed in Table 2.

The mean Early-Middle Silurian direction is $D=45.9°$, $I=-8.3°$, with $k=37.8$ and $A_{95}=8.5°$ (Fig. 3). The mean Middle-Late Silurian direction is $D=19.1°$, $I=10.2°$, with $k=28.6$ and $A_{95}=9.2°$ (Table 1). Both reversed and normal directions were observed (Fig. 3).

The Devonian, although the youngest proved the most complex as this time interval corresponds to the period of major orogenic activity in the area. Thermal activity associated with folding resulted in a strong Devonian overprinting, with a direction co-linear with the primary direction. A primary direction for the Devonian is $D=320.9°$, $I=-24.4°$, with $k=11.2$ and $A_{95}=28.7°$ (Fig. 3; Table 1) was obtained. This compares with, and is supported by a strong Devonian mean secondary overprint direction obtained from both the Ordovician and Silurian Formations of $D=325.7°$, $I=-12.8°$, with $k=26.7$ and $A_{95}=9.5°$.

By eliminating the known present field and the primary direction, the remaining directions could be identified when compared with primary directions obtained from younger formations. In this way, the age of the secondary directions were obtained. For example, for the Ordovician Formations secondary directions isolated were, 1) a Silurian overprint (corresponding to minor folding during the Benambran Orogeny), 2) the strong Devonian overprint (major deformation during the Bowning Orogeny), 3) a poorly

Table 2. Palaeomagnetic data.

Formation	Mnemonic	N	Decl	Incl	k	A_{95}	Lat (°N)	Long (°E)	DP	DM
Corrected primary directions										
Early Ordovician										
Walli And.	WA	25	225.59	−52.81	24.2	6.01	−10.46	6.26	5.73	8.30
Mt Pleasant And.	MPA	5	38.16	54.98	50.1	10.91	−12.20	0.01	10.98	15.48
Middle-Late Ordovician										
Cliefden Cvs Ls.	CCL	68	259.04	−58.58	28.6	3.28	13.20	20.21	3.61	4.89
Malongulli Fm.	MG	61	256.85	−52.18	63.7	2.30	8.07	24.68	2.16	3.15
Angullong Tuff	AT	32	259.65	−51.41	21.1	5.66	9.62	26.65	5.22	7.69
Early-Middle Silurian										
Rockdale Fm.	RD	12	45.17	1.41	63.9	5.47	−35.51	29.38	2.74	5.47
Millambri Fm.	ML	24	40.62	−18.38	21.3	6.57	−45.62	36.45	3.55	6.83
Belubula Shale	BS	19	33.47	−10.57	52.4	4.68	−48.00	23.96	2.40	4.74
Narragnal Ls.	NL	19	52.52	−1.43	174.8	2.55	−31.25	37.14	1.28	2.55
Middle-Late Silurian										
Canowindra Phy.	CP	9	21.46	15.71	20.7	11.57	−43.70	359.85	6.12	11.90
U. Avoca V. Sh.	UAVS	40	25.18	30.10	50.6	6.85	−34.74	358.64	4.22	7.60
Ghost Hill Fm.	GH	27	196.68	−22.77	37.2	4.62	−41.84	350.97	2.60	4.90
Mumbil Fm.	MB	78	188.69	1.18	10.5	5.20	−56.88	345.02	2.60	5.20
Early-Middle Devonian										
Dolerite intrus.	D	6	301.19	−34.02	103.6	6.61	−35.85	238.74	4.32	7.56
Cowra Granod.	CG	8	302.47	−31.62	30.2	10.24	−36.17	241.52	6.45	11.49
Cunningham Fm.	CH	17	312.38	8.10	52.9	4.95	−31.82	268.89	2.51	4.99
Tenandra Fm.	TF	11	154.78	41.87	26.7	9.00	−66.05	255.49	6.76	11.03
Secondary Components										
Silurian overprint										
Mt Pleasant And.	MPAO	5	233.43	6.00	18.4	18.32	−31.63	39.43	9.24	18.40
Devonian overprint										
Angullong Tuff.	ATO	11	328.06	−1.48	40.8	7.24	−45.55	279.26	3.62	7.24
Mt Pleasant And.	MPAO	6	325.14	−13.22	47.8	9.79	−48.00	271.00	5.09	9.99
Millambri Fm.	MLO	13	157.40	−27.01	9.2	14.45	−37.54	301.77	8.55	15.72
L. Avoca V. Sh.	LAVSO	14	316.63	−27.00	41.8	6.22	−46.34	254.23	3.68	6.76
Canomodine Ls.	CLO	13	326.40	−32.26	27.8	8.01	−55.94	258.36	5.09	9.03
Cliefden Cvs Ls.	CCLO	19	341.60	−19.47	31.2	6.11	−60.97	289.08	3.33	6.38
Ghost Hill Fm.	GHO	6	319.54	2.64	174.2	5.09	−38.37	272.98	2.55	5.09
Belubula Shale	BSO	24	320.00	−5.50	20.3	6.73	−41.60	269.66	3.39	6.75
Mesozoic? overprint										
Canomodine Ls.	CLO	7	14.74	−79.19	12.1	18.07	−53.47	140.01	32.69	34.37
Mt Pleasant And.	MPAO	17	29.32	−60.91	19.5	8.29	−65.53	87.50	9.70	12.68
Cugga Burga Vol.	CBVO	21	41.46	−75.75	32.0	5.70	−50.13	121.12	9.65	10.49
Narragnal Ls.	NLO	15	70.70	−77.80	18.6	9.12	−37.31	120.89	16.09	17.13
Tolga Calcaren.	TCO	46	357.09	−73.86	86.0	2.28	−63.62	152.06	3.70	4.11
Cunningham Fm.	CHO	18	7.27	−75.90	13.4	9.81	−59.06	142.68	16.65	18.08

defined Mesozoic overprint (Mesozoic activity associated with the break up of Australia and Antarctica during the Creatceous), and 4) the present field direction.

5. Discussion

The primary pole positions obtained are shown in Table 2 and these indicate a well defined apparent polar wander path for the Early Palaeozoic of South East Australia (Fig. 4). These new poles, when plotted with those used in the previous interpretations produce a stratigraphically coherent path (Fig. 5). The general trend of the Central Australian apparent polar wander path of McElhinny and Embleton (1974), defined by the Cambro-Ordovician pole positions LFG, JF, and DG to the Ordovician TS, AD, and the Siluro-Devonian MS pole position is the same as the trend defined by the new data (Fig. 5).

The new data for the Middle-Late Silurian however, indicate the start of a sharp leftwards swing of the path towards South America. The pole path then swings back

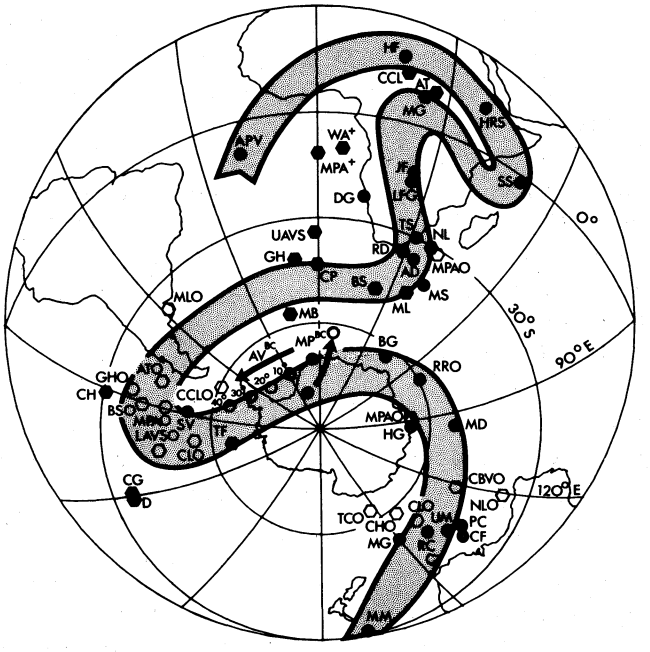


Fig. 5. Combining the mean of new poles for each time period and the other Australian poles, a single coherent path for Australia since the Late Precambrian is constructed. With this path three poles are in error, SV, AV, and MP. On bedding correction, the MP pole moves towards a stratigraphically correct position on the new path. The AV pole, however, moves towards the Devonian region of the path. This suggests both the SV and AV poles are secondary Devonian poles. See text for description. Solid circles are Australian poles listed in Fig. 1. Solid hexagons are new primary poles; open hexagons are new secondary poles (Mnemonic's are listed in Table 2). Open circles represent the bedding corrected Ainslie Volcanics (0–40° dip correction) and the Mugga Porphyry.

during the Early-Middle Devonian? (Fig. 5), corresponding with the path previously identified as representing the separate plate motions of South East Australia (MCELHINNY and EMBLETON, 1974).

If the ages of magnetization for the new Devonian pole are correct then the poles SV, MP, and AV plot out of sequence. Recent work in the Canberra area (M. Owen, personal communication) has shown that the Ainslie Volcanics (AV) and the Mugga Porphyry (MP) both previously thought to be flat lying, in fact have substantial dips. The Mugga Porphyry (MP) has an average south-westerly dip of 27° whereas the Ainslie Volcanics dip to the north-east at angles up to a maximum of 35°. Stratigraphic relations indicate the Ainslie Volcanics (old pole pos. 71°S, 353°E) are Late Wenlockian and not Early Devonian as previously documented, and therefore older than the earliest Ludlovian Mugga Porphyry (M. Owen, personal communication), (old pole pos. 80°S, 340°E). When these bedding corrections are applied, the Mugga Porphyry moves to the Middle-Late Silurian region (new pole pos. 63.2°S, 007.9°E; Table 1) of the new path (Fig. 5), but the Ainslie Volcanics move towards the Devonian region. The precise dip of the Ainslie Volcanics at the sampling sites is unknown, however dips range from horizontal to 35° about a strike of 179°, within the area sampled. Bedding corrected poles for a constant strike and varing dip range from 72.3°S, 341.2°E for 5° dip to 67.7°S, 285.3°E for a dip of 35° (Table 1). This suggests that the Ainslie Volcanics as measured by LUCK (1973) contain this strong Devonian overprint. The main folding occurred after the Devonian overprinting, during the Upper Devonian-Lower Carboniferous.

The Silurian Volcanics (SV) pole position lies on the new Devonian direction, suggesting this direction is in fact indicating the strong Devonian overprint direction found throughout the Ordovician and Silurian. Neither adequate cleaning nor has any fold test been applied on these samples. Further work on both the Ainslie Volcanics and the Silurian Volcanics using present techniques will prove or disprove this suggested explanation of the SV discrepency.

Small scale rotations between the tectonic provinces sampled and neighbouring tectonic provinces or the stable central Australian craton were not detectable, though geological evidence indicates small rotations did occur (SCHEIBNER, 1974). The 20° discrepency between the new South East Australian Ordovician pole and the central Australian Ordovician pole may indicate such tectonic movement.

I am grateful to Drs. M. W. McElhinny and C. T. Klootwijk for helpful discussion during the various stages of the work and to Mr. D. J. Edwards who assisted greatly in the operation and maintenance of the instruments. All are members of the Australian National University. My thanks also to Drs. M. Owen and G. A. M. Henderson of the Bureau of Mineral Resourses, Canberra, Australia, for valuable discussion concerning the age, structure, and geology of the Ainslie Volcanics and the Mugga Porphyry. Critical reading of the manuscript by M. W. McElhinny, C. T. Klootwijk, B. J. J. Embleton, and P. W. Schmidt was greatly appreciated.

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